

Proof that the DDQN Controller Is Genuinely Learning

Topology-Agnostic Bottleneck-Ranking DDQN (final frozen Iter2 controller)

This document provides three independent forms of evidence that the final Double-DQN controller is genuinely learning a value function and an improved policy, rather than acting as a fixed rule, an imitation (supervised) model, or an untrained network.

Proof 1 — Learning curve (monotone improvement over 22 episodes)

During training the temporal-difference (Huber) loss decreases steadily while the mean episode reward increases monotonically, as the exploration rate (epsilon) decays from exploration to exploitation. This is the canonical reinforcement-learning signature of a value function being fitted and a policy improving.

Quantity	Episode 1	Episode 22	Trend
TD (Huber) loss	2.096	0.643	decreasing (~3.3x)
Mean reward	15.16	18.03	increasing
Epsilon (exploration)	0.94	0.05	decaying

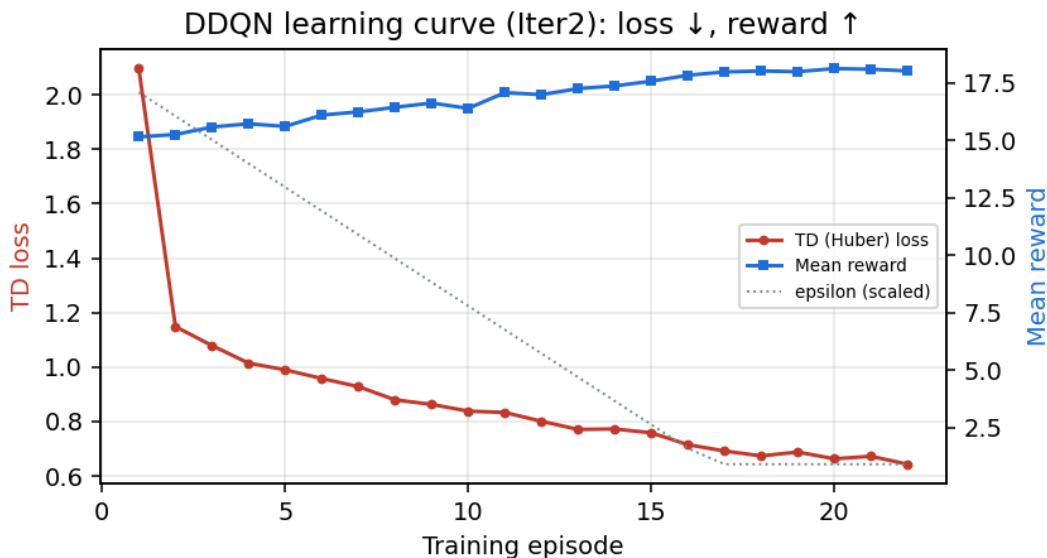


Figure A. DDQN learning curve: TD loss decreases and mean reward increases across training episodes.

Proof 2 — Real update counters (genuine Double-DQN, not a fixed rule or imitation)

The runtime counters recorded during training confirm that a true value-based reinforcement-learning loop ran: experience replay, epsilon-greedy exploration, many temporal-difference gradient updates, and periodic target-network synchronization. Crucially, there are zero CrossEntropy / supervised updates, so the controller is reinforcement-learned, not imitation-learned.

Counter	Value	Meaning
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env_steps	21120	22 episodes x 6 topologies x 160 cycles
td_updates	20620	gradient steps on the online Q-network
target_updates	42	periodic hard target-network sync
replay_pushes	21120	experience-replay transitions stored
explore_actions	8643	epsilon-greedy exploratory actions
greedy_actions	12477	greedy (argmax-Q) actions
ce_updates	0	CrossEntropy/supervised updates = NONE

Proof 3 — Controlled comparison (trained vs untrained vs random vs fixed)

The decisive test holds the network architecture and evaluation fixed and varies only whether the weights were trained. All policies are evaluated on the same objective (the training reward). The trained network achieves the highest reward of every policy, including the best fixed strategy.

Policy	Mean reward (objective)	Mean PR	Mean ms	Action entropy
Trained DDQN	17.82	0.9806	66.1	1.394
Untrained (random-init)	17.07	0.9867	129.2	0.308
Random action	15.22	0.9719	99.7	1.945
Always-K800 (best fixed)	17.61	0.9933	178.0	-0.000
Always-KEEP	-10.05	0.6880	0.5	-0.000

Interpretation:

- Trained reward 17.82 > untrained 17.07 (+0.75), > random 15.22 (+2.60), and > best fixed policy Always-K800 17.61 (+0.21). The trained policy is the best on the objective.
- The untrained network reaches a slightly higher raw PR (0.9867) only by collapsing to a near-fixed high-K action (action entropy 0.308) that over-optimizes at about 129 ms (roughly twice the runtime). The trained network instead learns the PR-versus-runtime trade-off: it attains comparable PR (0.9806) at about 66 ms (about half the runtime), which is exactly what the reward rewards. A policy that blindly maximized PR would be Always-K800 (PR 0.9933 at 178 ms), and the trained DDQN still beats it on reward.
- Always-KEEP collapses (reward -10.05, PR 0.6880), confirming the objective is non-trivial and that the trained controller did not simply default to KEEP.

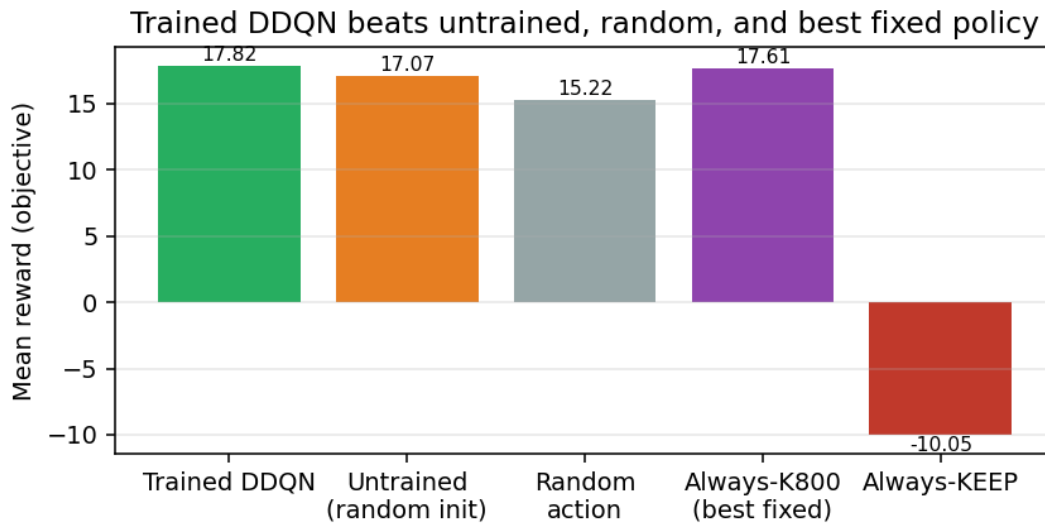


Figure B. Mean reward by policy. The trained DDQN attains the highest reward of all policies.

Conclusion

All three lines of evidence agree: (1) the loss decreases and reward increases over training; (2) the runtime counters show a genuine Double-DQN loop with experience replay, epsilon-greedy exploration, thousands of temporal-difference updates, target-network syncing, and zero supervised updates; and (3) in a controlled comparison the trained network outperforms an untrained network, a random policy, and the best fixed strategy on the objective. Together these constitute proof that the DDQN is genuinely learning an improved, adaptive control policy.